



Examiners' Report Summer 2026

Pearson Edexcel GCE

In Further Mathematics (9FM0)

Paper 4B Further Statistics 2

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General comments

The paper gave learners a range of opportunities to demonstrate their understanding of this part of the specification. Many responses showed secure understanding of the statistical techniques required, with learners generally able to select and apply more advanced methods across a range of scenarios.

Time did not appear to be a significant issue, with responses generally engaging with all questions on the paper. The more demanding parts were Q4(b) and Q7(c). There was evidence that learners engaged with the final part, and a number of responses gained substantial credit.

Comments on individual questions

Question 1

This question tested understanding of correlation and learners were able to make a good start to the paper.

Part (a)

In part (a), learners were generally able to find the value of the PMCC successfully. Marks were occasionally not awarded, either when responses did not recognise the need to find S_{yy} or where copying errors were made.

Part (b)

Part (b) was generally answered successfully. Some responses correctly referred to the positive correlation found in part (a), but did not complete the interpretation by stating whether this supported the use of a linear model for these data.

Part (c)

In part (c), most learners recognised that the gradient of the linear model was required. Some responses were less successful in using the value of 2 given in the question to determine the corresponding quantity of nitrogen.

Part (d)

In part (d), some responses did not explain or recognise that the linear model should minimise the RSS, which was required for full credit. Other responses stated only the formula for RSS, without substituting their values, and then gave the value 2.15. As this value was already given in the table, further working was needed to demonstrate the method.

Question 2

This question focused on a hypothesis test for the difference between the means of two Normal distributions, and a test for whether two independent random samples came from Normal populations with equal variances. It was generally answered well.

Part (a)

Part (a) required learners to write down hypotheses which was generally done well using the correct notation.

Part (b)

Part (b) was similarly answered successfully with responses showing correctly how to calculate a pooled estimate of the variance.

Part (c)

In part (c), responses generally showed secure understanding of the hypothesis test. Where marks were not gained, this was often because the conclusion was either incorrect or not sufficiently linked to the context of the question.

Part (d)

A similar issue was seen in part (d), where some responses did not give a fully contextualised conclusion. For example, some stated only that there was a “different variance”, which did not provide enough information. In some cases, responses used the sample standard deviations rather than the sample variances when finding the test statistic.

Part (e)

In part (e), some responses included excellent explanations. Many responses named the test used in part (c) and explained why unequal variances made it unreliable. Other responses gave brief but correct statements, while some incorrectly stated that the test was reliable or focused on the means rather than the variances.

Question 3

This question tested Spearman's rank correlation and carrying out a hypothesis that a correlation is zero.

Part (a)

Part (a) proved more demanding. Responses were often successful in ranking both sets of data, including the use of tied ranks.

However, when tied ranks are present, Spearman's rank correlation coefficient should be calculated by applying the PMCC formula to the ranks. Many responses instead used the standard Spearman's rank correlation formula,

$$r_s = 1 - \frac{6\sum d^2}{n(n^2 - 1)}, \text{ despite the presence of tied ranks.}$$

Part (b)

Part (b) was answered more successfully than part (a). Many responses stated the hypotheses and found the critical value for the test. However, stating the critical region proved more demanding, with some responses showing only one region, not expressing the region in terms of r , or not providing a critical region.

Part (c)

Part (c) was answered well, particularly where follow-through from parts (a) and (b) was applied correctly. Where credit was not gained, this was typically due to insufficient context in the conclusion.

Question 4

This question assessed understanding of the continuous uniform distribution. Learners usually made good progress in parts (a), (c) and (e), while parts (b) and (d) required a fuller understanding of the steps involved.

Part (a)

Part (a) was answered correctly by almost all candidates, which did not require the fraction to be in simplest form. A rounded value was not acceptable.

Part (b)

In part (b), many responses did not recognise that the setup meant the condition was only possible at one end of the distribution. Some reached the value $k = 7.2$ and treated this as the final answer, rather than using it to complete the required calculation. Other responses began with a sketch, but assumed that a probability of 0.1 should be placed in each of two regions.

Part (c)

Part (c) was typically answered correctly. Where credit was not gained, this was usually due to occasional manipulation errors.

Part (d)

In part (d), the most common incorrect answer was a confidence interval with both limits 3 units too low. In addition, responses did not always use a z -value to the required accuracy (2.3263). Learners should be reminded to use the full value shown in the formula booklet, or sufficiently accurate values from their calculator.

Part (e)

Most learners attempted part (e), and many were able to gain some credit. Where full credit was not gained, this was typically due to the use of incorrect terminology, for example referring to the significance level rather than the confidence level. Other responses stated that more samples should be taken, rather than indicating that the sample size should be increased.

Question 5

This question assessed confidence intervals for the mean and the variance. Part (c) required a fuller understanding of confidence intervals, and errors from earlier parts often meant that full credit was not gained.

Part (a)

In part (a) the common issue seen in responses was in the construction of the confidence interval where several learners used the Normal distribution instead of the t – distribution.

Part (b)

In part (b), responses typically stated the correct critical values, but in calculating the lower and upper limits responses were seen occasionally using the sample standard deviation instead of the sample variance or an incorrect value for n .

Part (c)

In part (c), learners were asked for the smallest estimate, but several responses did not recognise that this required using the largest possible values of both the mean and the variance. Some responses used the midpoints of the intervals, while others used a combination of the lower limit from one interval and the upper limit from the other.

Question 6

This question required learners to use integration confidently to find probabilities for a continuous random variable. Responses that recognised the integral as a standard form were generally successful. Some responses expanded the denominator, but this approach did not generally lead to a successful solution.

Part (a)

In part (a) most learners mentioned symmetry, with excellent explanations explaining how this made $\text{mean} = \text{mode} = \text{median}$. Some responses incorrectly stated 1, as it was the highest point.

Part (b)

In part (b), learners who spotted the integration to $\arctan$ usually scored full marks, whereas some learners attempted to use partial fractions, which was not making progress successfully.

Part (c)

In part (c), some learners attempted to find regions such as $P(1 < x < 1.5)$ as their final answer. Learners who correctly wrote out the conditional probability formula usually went on to get full marks.

Part (d)

In part (d), most responses correctly placed the highest point to the left of $x = 1$. However, many responses did not show the correct shape of the graph, with some showing a negative quadratic shape or a linear section to the right. Some responses also did not show $x = 2$ at a lower point than $x = 0$. Learners should be encouraged to consider both the sketch of the original graph and the given function in such cases. Considering the shape of a reciprocal function would have supported a more accurate sketch in this situation.

Question 7

This question assessed combinations of independent Normal random variables. Most learners made good progress in the first two parts. The final part was one of the most demanding on the paper, although some correct responses were seen.

Part (a)

Part (a) was generally well answered, with most responses finding both the mean and variance correctly.

Part (b)

Part (b) was also answered well. Some responses avoided standardising by rearranging the inequality and then using the inverse Normal function on their calculator to find the critical value. Occasional arithmetical slips were seen when calculating the variance or when taking the square root during standardisation.

Part (c)

Part (c) was particularly demanding. Responses often did not expand the brackets and simplify the expression to obtain the correct model. Few responses showed recognition that simplifying the expression for T resulted in the two J_2 terms cancelling, leaving $(m - 2)J_1$. Some responses obtained the correct final answer from a variance based on incorrect working, where not all of the individual variances had been added and some had been subtracted. In these cases, the correct answer was not supported by a valid method.